

SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Computer Vision with OpenCV

COURSE CODE: DSA0201

COURSE OUTCOME COVERED

***CO4:** Evaluate recognition and learning methods including machine learning and deep learning models (CNNs, GANs, SVM, HMM, etc.) for vision tasks.. (BL5)*

Assessment Tool Weightage: 50%

QUESTIONS: 5

Total Marks:50

Annexure A

Data Interpretation

Code of Conduct:

I B.Deepthi-(192424269) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: B.Deepthi

Q1. Hough Transform & Shape Detection

Scenario

The objective is to detect **roads, rivers and canals** as linear features in noisy satellite images. The images may contain clouds, shadows, intersections and partially hidden lines.

Step 1: Understand the Data

- Input is a **1024 × 1024 grayscale satellite image**.
- Noise may occur due to **clouds and shadows**.
- Several lines may appear at the same time.
- Roads and rivers can intersect or be partially occluded.
- Therefore, detecting straight or approximately straight edges is required.

Step 2: Analysis

The **Hough Transform** converts edge points from image space into parameter space.

For a line:

$$[x\cos\theta + y\sin\theta = \rho]$$

where:

- (ρ) = distance from origin
- (θ) = angle of the line

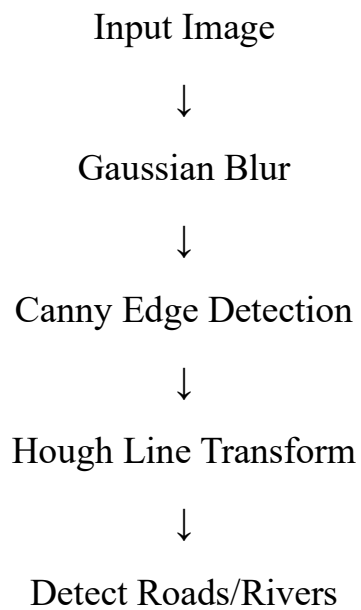
Each detected edge point votes for possible $((\rho, \theta))$ values. A strong peak indicates a possible line.

Step 3: Application

A suitable processing pipeline is:

Satellite Image → Grayscale → Noise Removal → Edge Detection → Hough Transform → Detected Lines

Example:



For OpenCV, `cv2.HoughLines()` or `cv2.HoughLinesP()` can be used.

Step 4: Conclusion

Hough Transform is effective for identifying **linear structures** even when some portions of the line are missing. However, heavy clouds and shadows can create false edges, so preprocessing is important.

Step 5: Clarity

Final result: The system detects the location, orientation and length of major linear features in satellite images.

Q2. PCA & Shape Priors for Object Recognition

Scenario

The task is to segment organs such as the **liver or heart** from MRI scans using a PCA-based shape model. The training set contains **50–100 annotated organ masks**, while the test set contains MRI images with noise or incomplete boundaries.

Step 1: Understand the Data

- Input: **512 × 512 grayscale MRI images**.
- Training images contain manually annotated organ masks.
- PCA is used to learn common organ shapes.
- Test images may contain noise and missing boundaries.

Step 2: Analysis

Each organ shape is represented using its boundary points.

Let the mean shape be:

$$[\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i]$$

The covariance matrix is calculated and PCA finds its principal components.

A new shape can be represented as:

$$[x = \bar{x} + Pb]$$

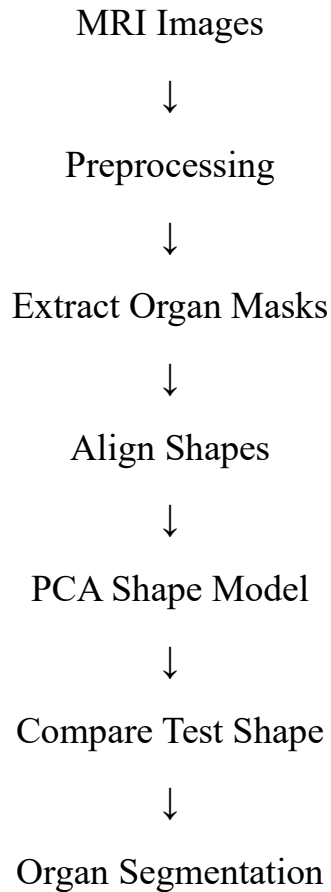
where:

- $(\bar{x})$ = mean organ shape
- (P) = principal components
- (b) = shape parameters

Thus, PCA captures the **most important variations in organ shape**.

Step 3: Application

Processing:



The learned shape prior helps identify an organ even when its boundary is incomplete.

Step 4: Conclusion

PCA reduces the dimensionality of organ shapes and provides a compact representation. Shape priors improve segmentation when MRI images contain **noise or missing boundaries**.

Step 5: Clarity

Final result: The system uses the learned average organ shape and its variations to obtain a more reliable segmentation.

Q3. Machine Learning Methods: SVM, HMM, GMM and EM

Scenario

The objective is to recognize human activities such as **walking, running and jumping** from video feature trajectories. The input contains joint positions or optical-flow vectors over time.

Step 1: Understand the Data

- Videos are recorded at **30–60 FPS**.
- Features include joint coordinates ((x,y)) or optical flow.
- Each video has an activity label.
- The data has a strong **time-dependent relationship**.

Step 2: Analysis

Different algorithms have different roles:

Method Main Purpose

SVM Classifies extracted features

HMM Models activities as sequences

GMM Represents feature distributions

EM Estimates hidden GMM parameters

For example, an activity can be represented as:

$$[X=(x_1, x_2, \dots, x_T)]$$

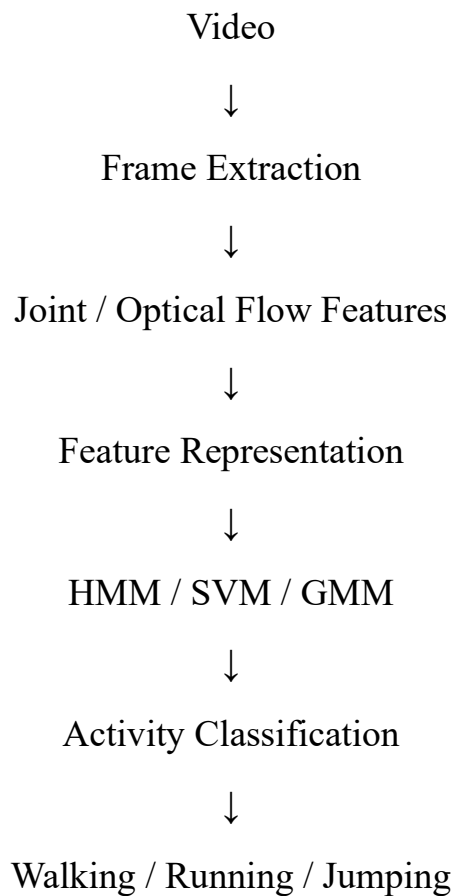
where each (x_t) represents the features at time (t) .

HMM is especially useful because:

$$[P(X,S)=P(S_1)\prod_{t=2}^T P(S_t|S_{t-1})\prod_{t=1}^T P(X_t|S_t)]$$

It models how one activity state changes into another over time.

Step 3: Application



For example, repeated leg and arm movement patterns can be classified as **walking**.

Step 4: Conclusion

SVM provides strong classification, while HMM is better suited for **temporal activity sequences**. GMM and EM can model complex feature distributions. Therefore, combining temporal and statistical models can improve recognition.

Step 5: Clarity

Final result: The system identifies the human activity by learning spatial features and their changes over time.

Q4. Deep Learning: CNNs, Transformers and GANs

Scenario

The objective is to detect **rare obstacles** such as traffic cones and animals in self-driving car images. The dataset contains only **500–1000 labeled images**, with optional GAN-generated images for augmentation.

Step 1: Understand the Data

Input consists of RGB urban street images.

- Images are approximately **1024 × 768 pixels**.
- Bounding boxes identify obstacles.
- The dataset is relatively small.
- Rare objects may not appear frequently.

Step 2: Analysis

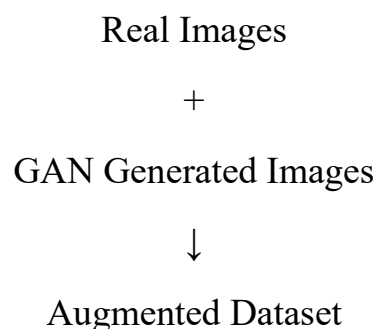
CNN: Extracts spatial features such as edges, shapes and object patterns.

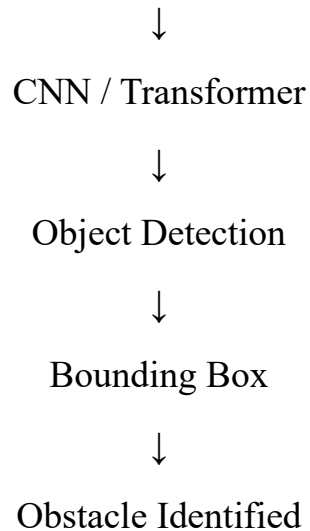
Transformer: Uses attention to understand relationships between different regions of the image.

GAN: Generates realistic synthetic images to increase training data.

GAN augmentation can help overcome the problem of insufficient rare-object examples.

Step 3: Application





For example, if an animal appears on the road, the detector should identify its location using a bounding box.

Step 4: Conclusion

CNNs are effective for visual feature extraction, Transformers can capture wider image relationships, and GANs can increase the diversity of training samples. For a small dataset, **data augmentation is particularly important**.

Step 5: Clarity

Final result: The proposed system improves rare-obstacle detection by combining deep visual learning with synthetic data generation.

Q5. Graph-Based Learning for Shape Correspondence

Scenario

The objective is to find corresponding points between **3D human face meshes** having different numbers of vertices. The training set contains annotated landmark correspondences.

Step 1: Understand the Data

Input consists of 3D face meshes.

- Meshes contain approximately **5,000–50,000 vertices**.
- Different faces can have different vertex counts.
- Training data contains landmark correspondences.
- Graph adjacency matrices can represent connections between vertices.

Step 2: Analysis

A 3D face can be represented as a graph:

$$[G=(V,E)]$$

where:

- (V) = vertices of the face
- (E) = connections between neighboring vertices

Important facial landmarks include:

Forehead



Eye

Eye



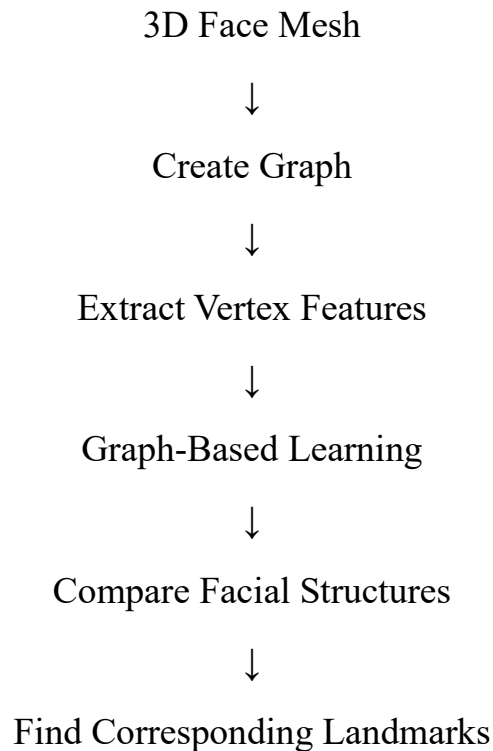
Nose



Mouth

Graph-based learning considers both **vertex features and relationships between neighboring vertices**.

Step 3: Application



For example, the nose-tip point in one face can be matched with the corresponding nose-tip point in another face.

Step 4: Conclusion

Graph-based learning is suitable because a 3D face is naturally represented as interconnected points. It can handle differences in mesh structure and identify corresponding facial landmarks.

Step 5: Clarity

Final result: The system establishes meaningful point-to-point correspondence between different 3D face meshes.